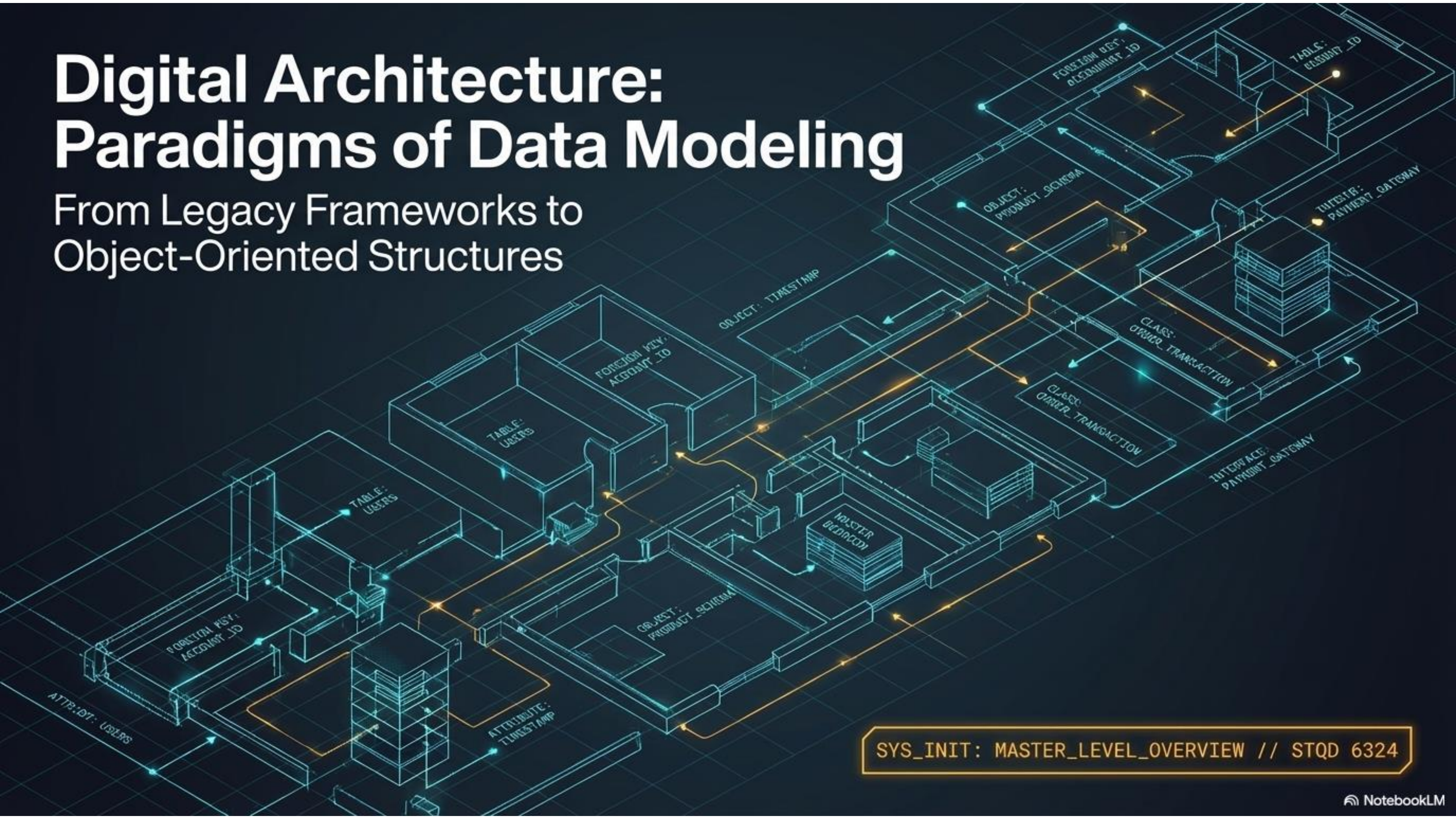


# From Legacy Frameworks to Object-Oriented Structures

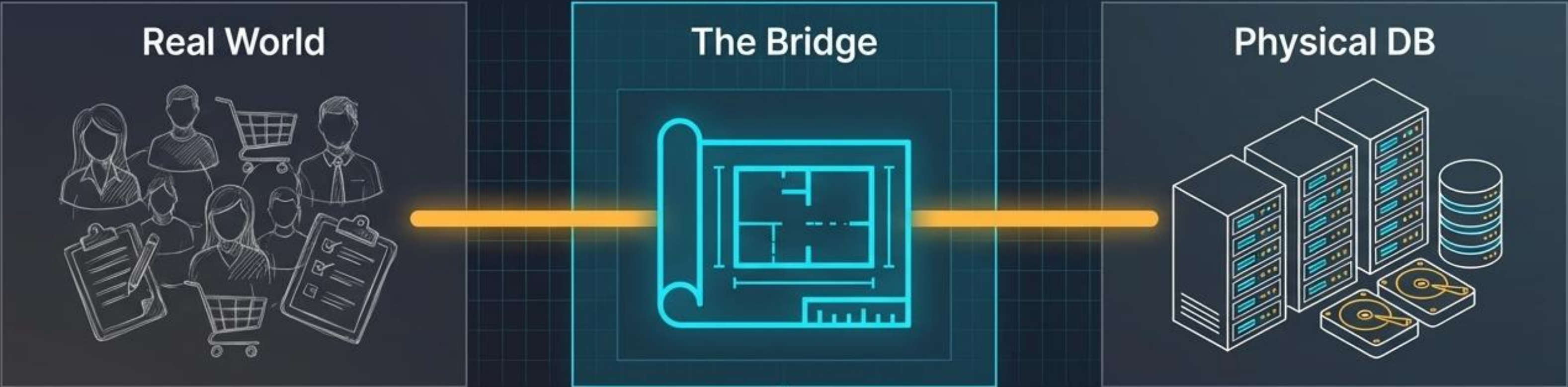


SYS\_INIT: MASTER\_LEVEL\_OVERVIEW // STQD 6324



# The Conceptual Bridge

A data model acts as the architectural blueprint between real-world events and physical hardware storage. It represents data exclusively as the user sees it, rendering it entirely independent of underlying hardware or software constraints.



## 01. ERD (Entity-Relationship Diagram)

The visual communication tool for stakeholders.

## 02. Data Dictionary

The comprehensive metadata document (detailing objects, types, rules) required by developers to construct the physical database.

| Column      | Data Type    | Description                            |
|-------------|--------------|----------------------------------------|
| employee_id | int          | Primary key of a table                 |
| first_name  | nvarchar(50) | Employee first name                    |
| last_name   | nvarchar(50) | Employee last name                     |
| nin         | nvarchar(15) | National Identification Number         |
| position    | nvarchar(50) | Current position title, e.g. Secretary |
| dept_id     | Int          | Employee department. Ref. Departments  |



# The Architect's Toolkit



## Entity

A distinguishable, unique object in the real world used to collect and store data.

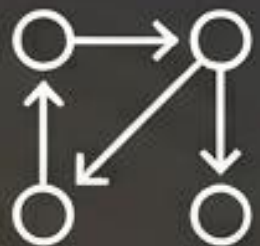
TABLE: CUSTOMER\_RECORD



## Attribute

A characteristic or descriptive property of a specific entity.

VARCHAR: last\_name



## Relationship

The structural association among entities. Expressed strictly as 1:1, 1:M, or M:N.

CONNECTIVITY: 1..\*



## Constraint

A strict rule ensuring logical data integrity across the model.

CHECK (GPA BETWEEN 0.00 AND 4.00)



# Legacy Constraints: The Pre-Relational Era



# Hierarchical Model



# Network Model

## Structure: Ordered Tree (Top-Down)

## Structure: Complex Graph

Navigation: “Left-list” path traversal; relies on physical parent-to-child proximity to minimize disk I/O time.

Navigation: Utilizes “Sets” containing at least two record types (Owner and Member).

**Capacity:** Strictly limited to 1:M relationships. Uses “Virtual Records” (logical pointers) to mitigate extreme data replication.

**Capacity:** Handles complex M:N relationships; a member can have multiple owners.

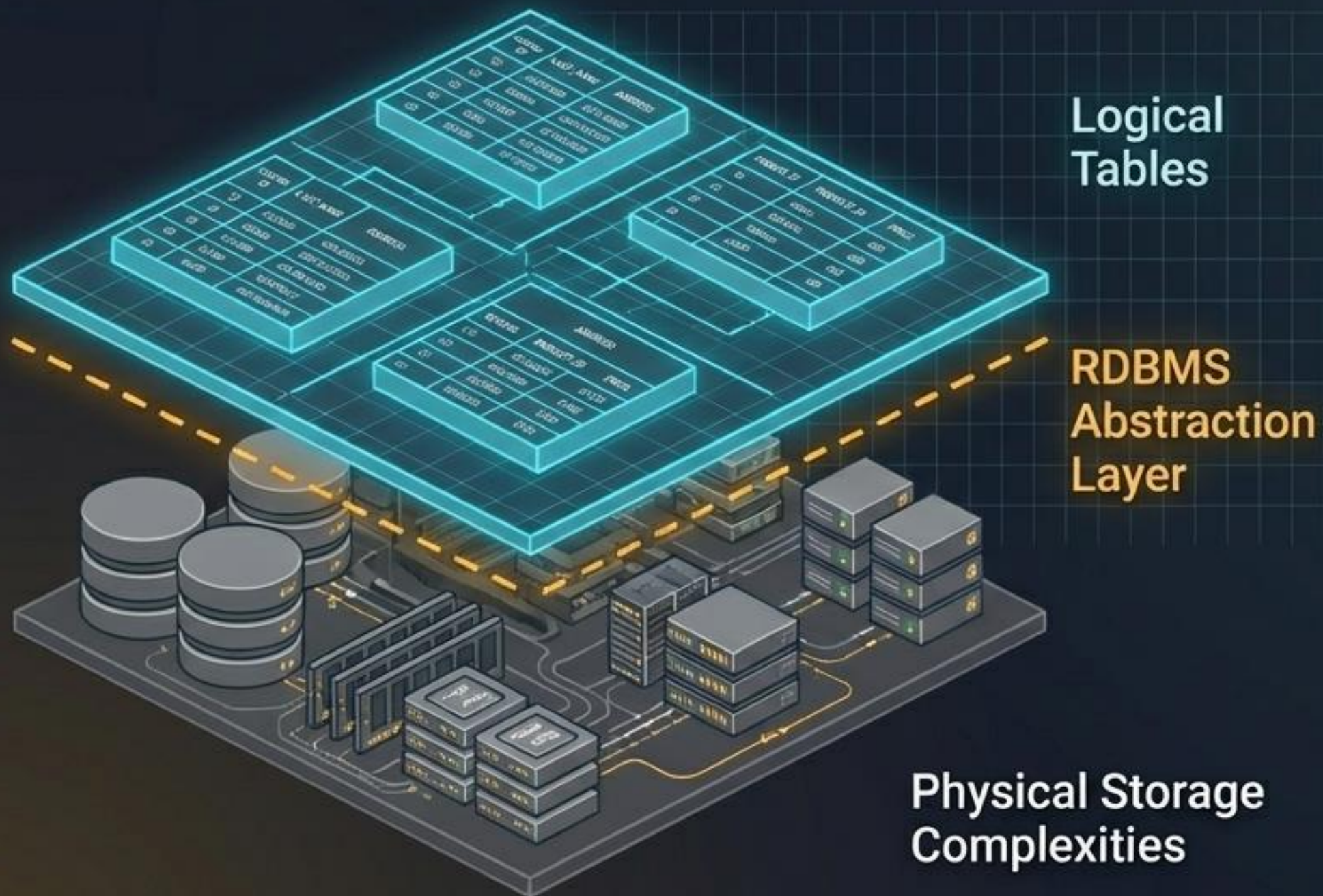
**Fatal Flaw:** Deletion anomalies (deleting a parent automatically cascades to child) and complete lack of structural independence.

**Fatal Flaw:** Overwhelming system complexity and an inflexible, pointer-based architecture that makes reorganization exceptionally difficult.



# The Relational Paradigm Shift (1970)

Developed by E.F. Codd, the relational model initiated a database revolution by achieving true Structural Independence.



## Core Mechanic

The Relational Database Management System (RDBMS) entirely absorbs the complex physical details of storage.

## The Result

Designers and users operate exclusively in a human-logical environment, interacting with data conceptually rather than mechanically via Ad Hoc Queries (SQL).



# Overcoming Islands of Information

primary key

Relational table

| AGENT      |             |             |               |                |              |
|------------|-------------|-------------|---------------|----------------|--------------|
| AGENT_CODE | AGENT_LNAME | AGENT_FNAME | AGENT_INITIAL | AGENT_AREACODE | AGENT_PHONE  |
| S01        | Anneson     | L'won       | A             | 317            | 990-888-8800 |
| S02        | Manly       | John        | B             | 320            | 958-808-0300 |
| S01        | Barward     | Maria       | N             | 317            | 996-864-8370 |
| S04        | Jarrson     | Canry       | A             | 330            | 981-437-3750 |
| S01        | Jomirman    | Davia       | A             | 332            | 958-334-3750 |

Link through AGENT\_CODE

foreign key

| CUSTOMER |           |           |             |              |              |                |            |
|----------|-----------|-----------|-------------|--------------|--------------|----------------|------------|
| CUS_CODE | CUS_LNAME | CUS_FNAME | CUS_INITIAL | CUS_AREACODE | CUS_PHONE    | CUS_RENEW_DATE | AGENT_CODE |
| S01      | Sursson   | Kum       | A           | 372          | 899-482-8880 | 12/28/2021     | S01        |
| S02      | Buntamd   | Kommy     | H           | 329          | 993-324-0320 | 12/21/2021     | S02        |
| S03      | Monson    | Bemon     | F           | 290          | 948-378-8896 | 09/24/2021     | S02        |

## Concept:

Relational schemas link independent data matrices (tables) by sharing common entity characteristics (Keys).

## The Problem Addressed:

Prevents isolated islands of data (e.g., Hospital Billing vs. Hospital Personnel systems) by weaving them into a unified, query-able architecture.

**WARNING:** Substantial hardware and system software overhead required compared to legacy pointer systems.



# The Drafting Language: Entity-Relationship Models

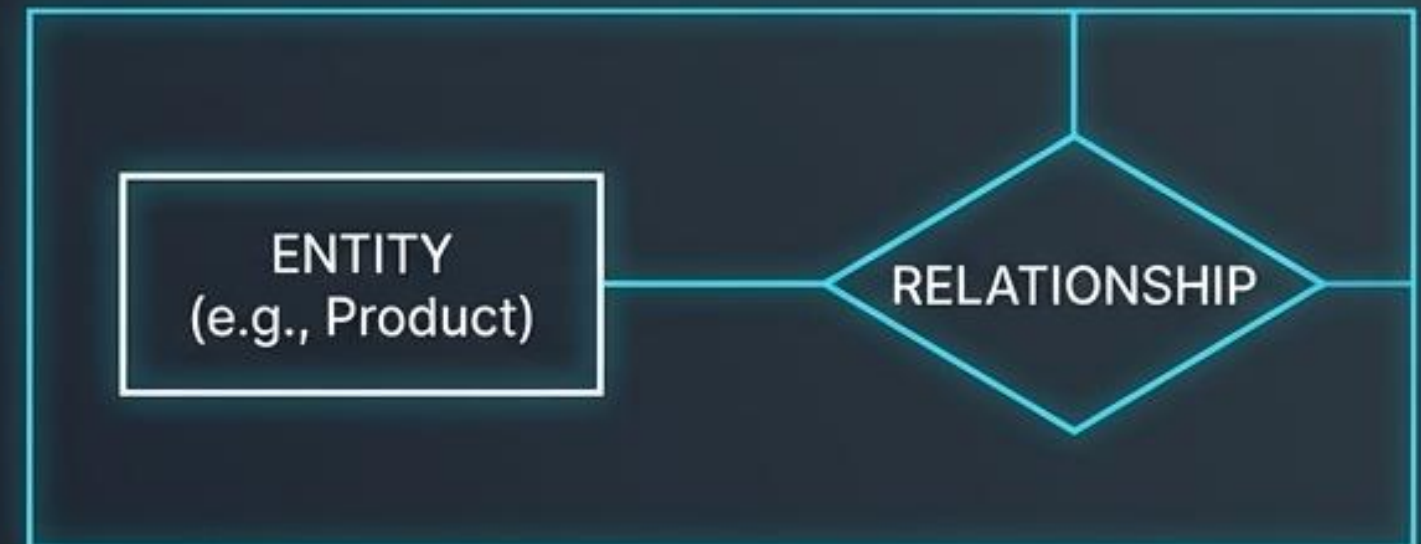
Introduced by Peter Chen in 1976, the ER Model provided a widely accepted graphical tool for data modeling, perfectly integrated with the relational paradigm.

## The Data

| ENTITY_ID | ENTITY_NAME | DESCRIPTION    |
|-----------|-------------|----------------|
| E001      | Product     | Inventory item |
| E002      | Order       | Customer order |
| E003      | Customer    | User profile   |



## The Draft



**Entity** → Maps to a full Relational Table.

**Entity Instance (Occurrence)** → Maps to a single Row.

**Connectivity** → Diamond or line endpoints denoting relationship types.



# The Notation Rosetta Stone: Chen vs. Crow's Foot

## Chen Notation

[1:M]: PAINTER paints PAINTING



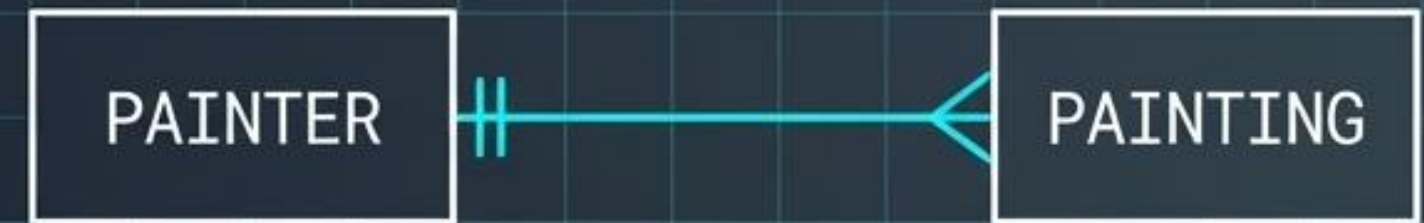
[M:N]: EMPLOYEE learns SKILL



[1:1]: EMPLOYEE manages STORE



## Crow's Foot Notation



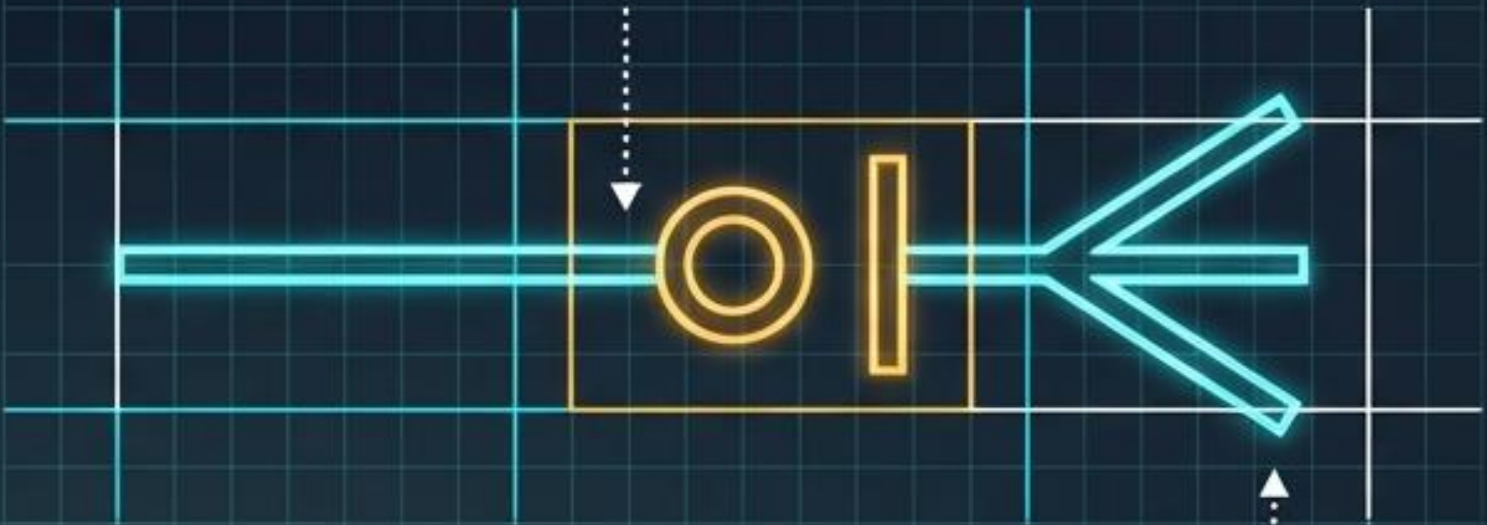


# Deciphering Crow's Foot Syntax

Cardinality constraints are embedded directly into the line endpoints, reading from the inside out.

**Inner Symbol (Minimums)**

Ring (O) = Minimum zero (Optional).  
Dash (|) = Minimum one (Mandatory).



**Outer Symbol (Maximums)**

Dash (|) = Maximum one.  
Foot (<) = Maximum many (infinite).

|  |                                       |
|--|---------------------------------------|
|  | Minimum zero, maximum one (optional)  |
|  | Minimum one, maximum one (mandatory)  |
|  | Minimum zero, maximum many (optional) |
|  | Minimum one, maximum many (mandatory) |



# Next-Generation Architecture: The Object-Oriented Shift

Based on the Semantic Data Model (SDM, 1981), OODM evolved to handle complex data types that traditional RDBMS struggled to manage.

## Relational Entity



Describes factual content and data only.

## Methods/Operations



## OO Object



An autonomous building block containing factual content, relationships, AND operations (behaviors) in a single encapsulated structure.

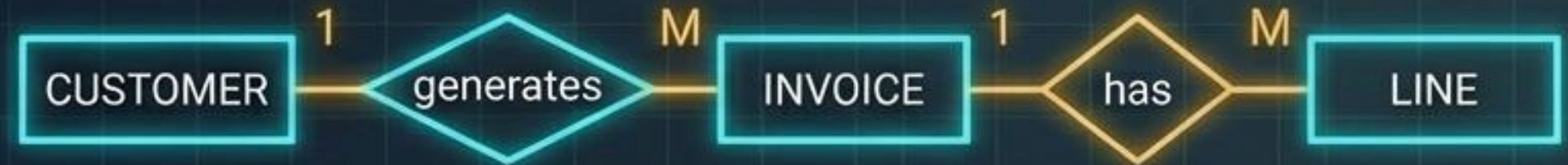


# Vertical Expansions: Classes & Inheritance

## OO Data Model



## ER Model



### The Mechanics:

**Classes:** Objects sharing similar characteristics are grouped into modular classes.

**Inheritance:** The architectural ability of an object to inherit attributes and methods from classes above it in the hierarchy.

### The ROI:

As transaction requirements grow sophisticated, OODM guarantees high data integrity, semantic richness, and crucial code reusability.

**ERROR\_TRACE:** High system overhead | Steep learning curve | Complex navigational access



# Master Architectural Synthesis

|                                        | Relational                                             | Entity-Relationship                              | Object-Oriented                                                          |
|----------------------------------------|--------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------|
| <b>Primary Building Block</b>          | The <b>Matrix Table</b><br>(Rows/Columns)              | The <b>Conceptual Entity</b>                     | The <b>Encapsulated Object</b>                                           |
| <b>Data &amp; Behavior Integration</b> | <b>Strictly Separated</b><br>(Logic vs. Storage)       | <b>Data only</b><br>(Visual Draft)               | <b>Fully Integrated</b><br>(Data + Methods)                              |
| <b>Primary Strength</b>                | <b>Structural Independence &amp; Ad Hoc SQL</b>        | Unmatched <b>Conceptual Simplicity</b>           | Code <b>Reusability &amp; Semantic Richness</b>                          |
| <b>Ideal Use Case</b>                  | Highly structured, standardized business intelligence. | Planning, communication, and visual translation. | Highly complex, interconnected, <b>behavior-driven</b> software systems. |